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1 C:\Users\13400f_RTX4060ti\AppData\Local\Programs\Python\
  Python313\python.exe "D:\MoleColyte\Training Scripts\train_Tox21.py"
2 Splitting: Train: 4408 |Val: 551 | Test: 551
3 Splitting: Train: 4408 |Val: 551 | Test: 551
4 Kiln: cuda
5 Scanning training data to calculate exact imbalance penalties...
6 Calculated Pathway Penalties: [33.4 49.1 8.6 30. 8.7 24.2 50.3 7.6 28.8
  22.3 7.7 20.5]
7
8 Starting Epoch 1
9   Epoch 1/20 | Batch 0 | Loss: 156.5921
10 ==> Epoch 1 Train Complete | Average BCE Loss: 6.6667
11   Validation Phase | Avg Val Loss: 1.3170
12   □ New best Tox21 model saved! (Lowest Val Loss: 1.3170)
13
14 Starting Epoch 2
15   Epoch 2/20 | Batch 0 | Loss: 1.0489
16 ==> Epoch 2 Train Complete | Average BCE Loss: 1.3064
17   Validation Phase | Avg Val Loss: 1.3166
18   □ New best Tox21 model saved! (Lowest Val Loss: 1.3166)
19
20 Starting Epoch 3
21   Epoch 3/20 | Batch 0 | Loss: 1.6793
22 ==> Epoch 3 Train Complete | Average BCE Loss: 1.3063
23   Validation Phase | Avg Val Loss: 1.3165
24   □ New best Tox21 model saved! (Lowest Val Loss: 1.3165)
25
26 Starting Epoch 4
27   Epoch 4/20 | Batch 0 | Loss: 1.3248
28 ==> Epoch 4 Train Complete | Average BCE Loss: 1.3056
29   Validation Phase | Avg Val Loss: 1.3163
30   □ New best Tox21 model saved! (Lowest Val Loss: 1.3163)
31
32 Starting Epoch 5
33   Epoch 5/20 | Batch 0 | Loss: 1.2140
34 ==> Epoch 5 Train Complete | Average BCE Loss: 1.3060
35   Validation Phase | Avg Val Loss: 1.3161
36   □ New best Tox21 model saved! (Lowest Val Loss: 1.3161)
37
38 Starting Epoch 6
39   Epoch 6/20 | Batch 0 | Loss: 1.2841
40 ==> Epoch 6 Train Complete | Average BCE Loss: 1.3051
41   Validation Phase | Avg Val Loss: 1.3160
42   □ New best Tox21 model saved! (Lowest Val Loss: 1.3160)
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43
44 **Starting Epoch 7**
45 **Epoch 7/20 | Batch 0 | Loss: 0.9588**
46 **==> Epoch 7 Train Complete | Average BCE Loss: 1.3058**
47 **Validation Phase | Avg Val Loss: 1.3161**
48 **Model did not improve. Best Val loss remains: 1.3160**
49
50 **Starting Epoch 8**
51 **Epoch 8/20 | Batch 0 | Loss: 1.1099**
52 **==> Epoch 8 Train Complete | Average BCE Loss: 1.3057**
53 **Validation Phase | Avg Val Loss: 1.3161**
54 **Model did not improve. Best Val loss remains: 1.3160**
55
56 **Starting Epoch 9**
57 **Epoch 9/20 | Batch 0 | Loss: 1.2302**
58 **==> Epoch 9 Train Complete | Average BCE Loss: 1.3052**
59 **Validation Phase | Avg Val Loss: 1.3160**
60 **□ New best Tox21 model saved! (Lowest Val Loss: 1.3160)**
61
62 **Starting Epoch 10**
63 **Epoch 10/20 | Batch 0 | Loss: 1.0737**
64 **==> Epoch 10 Train Complete | Average BCE Loss: 1.3048**
65 **Validation Phase | Avg Val Loss: 1.3160**
66 **□ New best Tox21 model saved! (Lowest Val Loss: 1.3160)**
67
68 **Starting Epoch 11**
69 **Epoch 11/20 | Batch 0 | Loss: 1.0296**
70 **==> Epoch 11 Train Complete | Average BCE Loss: 1.3059**
71 **Validation Phase | Avg Val Loss: 1.3161**
72 **Model did not improve. Best Val loss remains: 1.3160**
73
74 **Starting Epoch 12**
75 **Epoch 12/20 | Batch 0 | Loss: 1.0930**
76 **==> Epoch 12 Train Complete | Average BCE Loss: 1.3061**
77 **Validation Phase | Avg Val Loss: 1.3160**
78 **Model did not improve. Best Val loss remains: 1.3160**
79
80 **Starting Epoch 13**
81 **Epoch 13/20 | Batch 0 | Loss: 1.0079**
82 **==> Epoch 13 Train Complete | Average BCE Loss: 1.3050**
83 **Validation Phase | Avg Val Loss: 1.3161**
84 **Model did not improve. Best Val loss remains: 1.3160**
85
86 **Starting Epoch 14**

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87     Epoch 14/20 | Batch 0 | Loss: 1.4366
88 ==> Epoch 14 Train Complete | Average BCE Loss: 1.3055
89     Validation Phase | Avg Val Loss: 1.3160
90 Model did not improve. Best Val loss remains: 1.3160
91
92 Starting Epoch 15
93     Epoch 15/20 | Batch 0 | Loss: 1.2837
94 ==> Epoch 15 Train Complete | Average BCE Loss: 1.3058
95     Validation Phase | Avg Val Loss: 1.3160
96 Model did not improve. Best Val loss remains: 1.3160
97
98 Starting Epoch 16
99     Epoch 16/20 | Batch 0 | Loss: 1.4291
100 ==> Epoch 16 Train Complete | Average BCE Loss: 1.3052
101     Validation Phase | Avg Val Loss: 1.3160
102 Model did not improve. Best Val loss remains: 1.3160
103
104 Starting Epoch 17
105     Epoch 17/20 | Batch 0 | Loss: 1.6657
106 ==> Epoch 17 Train Complete | Average BCE Loss: 1.3059
107     Validation Phase | Avg Val Loss: 1.3162
108 Model did not improve. Best Val loss remains: 1.3160
109
110 Starting Epoch 18
111     Epoch 18/20 | Batch 0 | Loss: 1.5208
112 ==> Epoch 18 Train Complete | Average BCE Loss: 1.3048
113     Validation Phase | Avg Val Loss: 1.3160
114 Model did not improve. Best Val loss remains: 1.3160
115
116 Starting Epoch 19
117     Epoch 19/20 | Batch 0 | Loss: 1.2920
118 ==> Epoch 19 Train Complete | Average BCE Loss: 1.3057
119     Validation Phase | Avg Val Loss: 1.3160
120 Model did not improve. Best Val loss remains: 1.3160
121
122 Starting Epoch 20
123     Epoch 20/20 | Batch 0 | Loss: 1.2419
124 ==> Epoch 20 Train Complete | Average BCE Loss: 1.3049
125     Validation Phase | Avg Val Loss: 1.3162
126 Model did not improve. Best Val loss remains: 1.3160
127
128 Time: 0.29 minutes.
129
130 Process finished with exit code 0
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